2a	The total momentum before collision is <u>non-zero</u>. By COM, (total) <u>momentum is never zero</u>, so <u>not possible</u> for both blocks to be at rest simultaneously.	**A**	M1 A1

bi	By COLM, taking rightwards as positive (3M x 0.40) – (M x 0.25) = (3M x 0.20) + Mv v =0.35 m s^{-1}	**E**	 A1
bii	To right / away from block A, as direction to the right is taken as positive in **(b)(i)**	**E**	B1
c	relative speeds of approach is non-zero, and relative speeds of separation is zero Relative speed of approach is <u>not equal</u> the relative speed of separation, Hence <u>inelastic</u> collision.	**A**	M1 A1